

Department of Computer Science and Engineering
Islamic University of Technology (IUT)
A subsidiary organ of OIC

Lab Report 01

CSE 4412 : Computer Networks Lab

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Section: SWE
Semester: 4th
Academic Year: 2022-2023

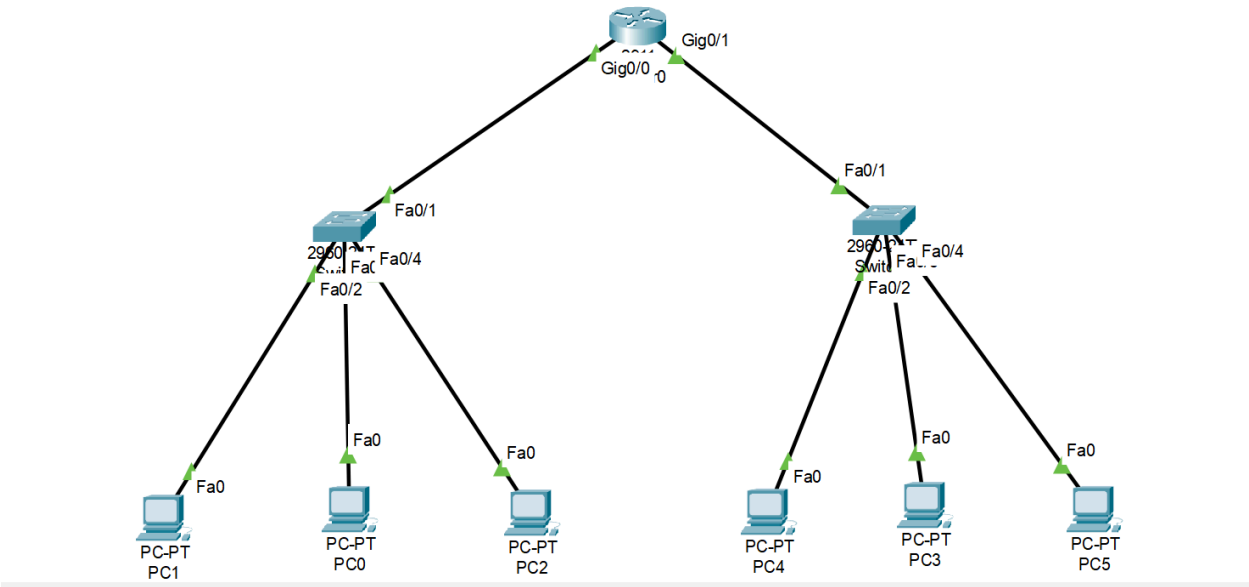
Date of Submission: 28/01/2024

Title: Configure router using static routing to connect multiple networks in Cisco Packet Tracer

Objectives:

1. Understand how to operate Cisco Packet Tracer
2. Learn to create and connect multiple networks using static routing
3. Understand wiring of different network components like router, switch, PC etc.
4. Configure router and switch interfaces
5. Verify connectivity of the network
6. Understand the basics of IP Subnetting
7. Learn to subnet a network following given specifications

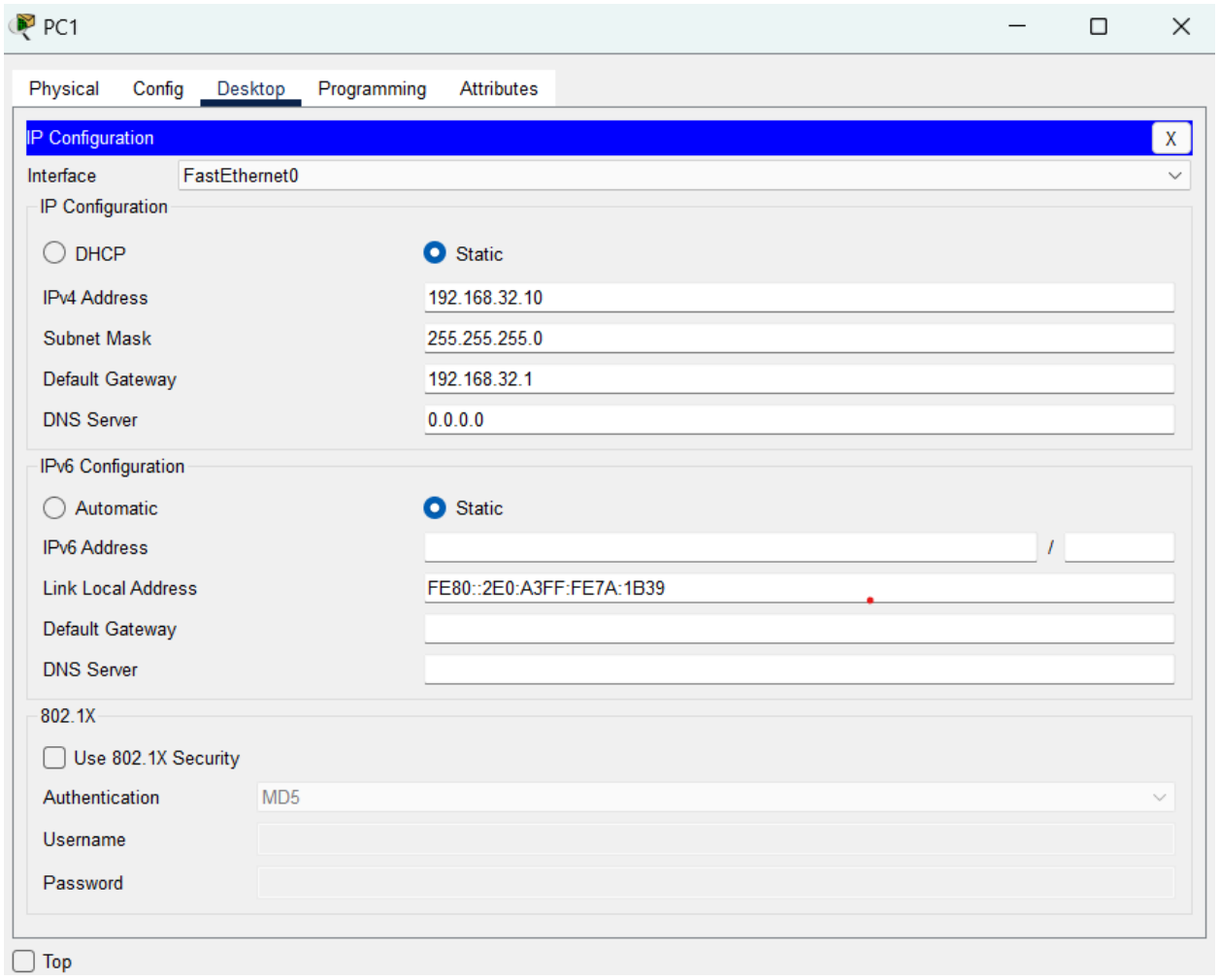
Diagram of the experiment:



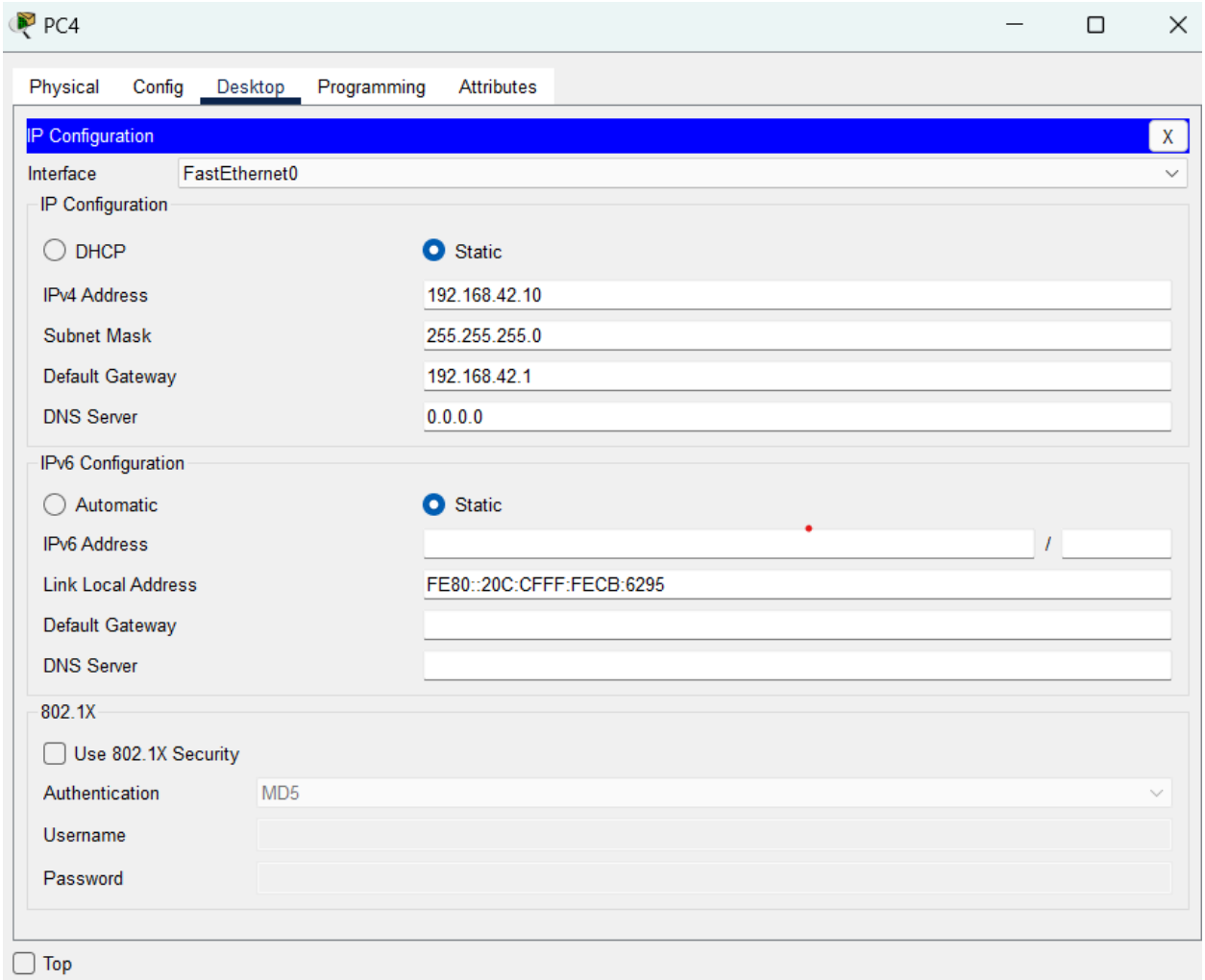
Working Procedure:

Step 1: Setting IP Address

I set the IP address of the PCs. For PC1,PC0,PC2 IP address was set to 192.168.32.10, 192.168.32.20, 192.168.32.30. The Default gateway for these 3 PCs are 192.168.32.1.



For PC4,PC3,PC5 IP address was set to 192.168.42.10, 192.168.42.20, 192.168.42.30. The Default gateway for these 3 PCs are 192.168.42.1.



Step 2: Renaming Host

To rename host of router, I clicked on the router, went to CLI and wrote the following code:

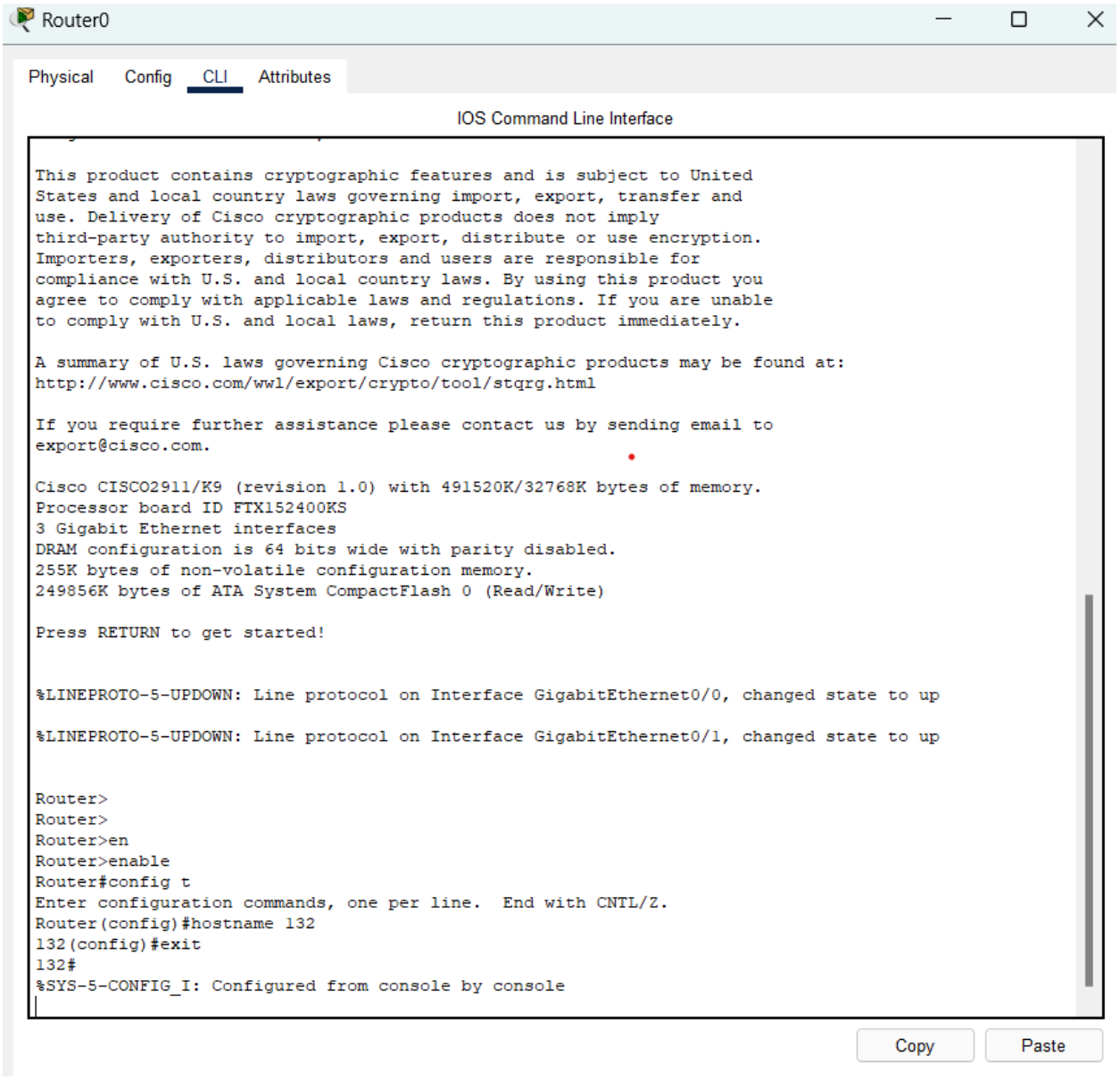
n

Router>enable

Router# config t

Router(config)# hostname 132

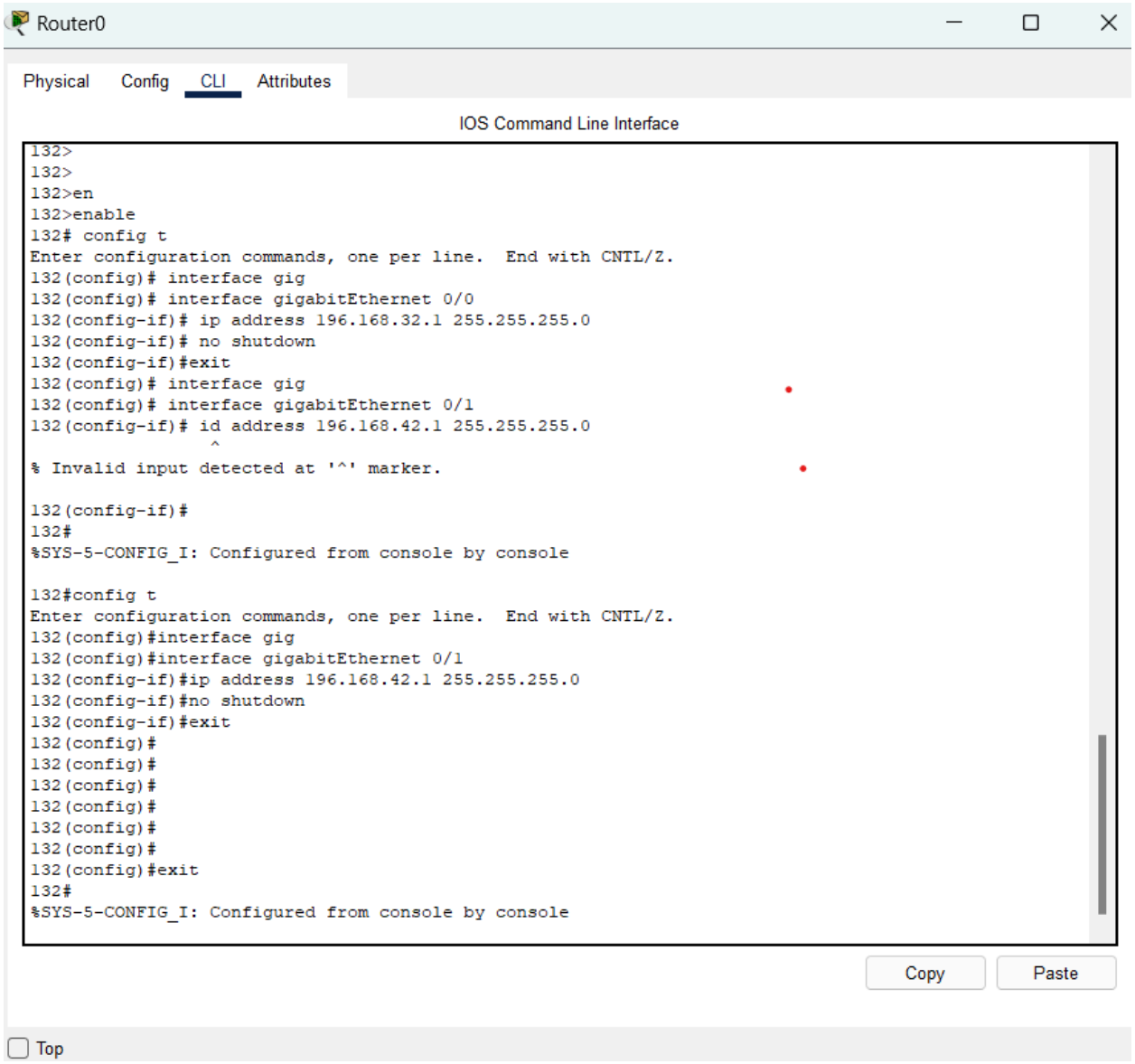
132(config)# exit



Step 3: Configuring Router

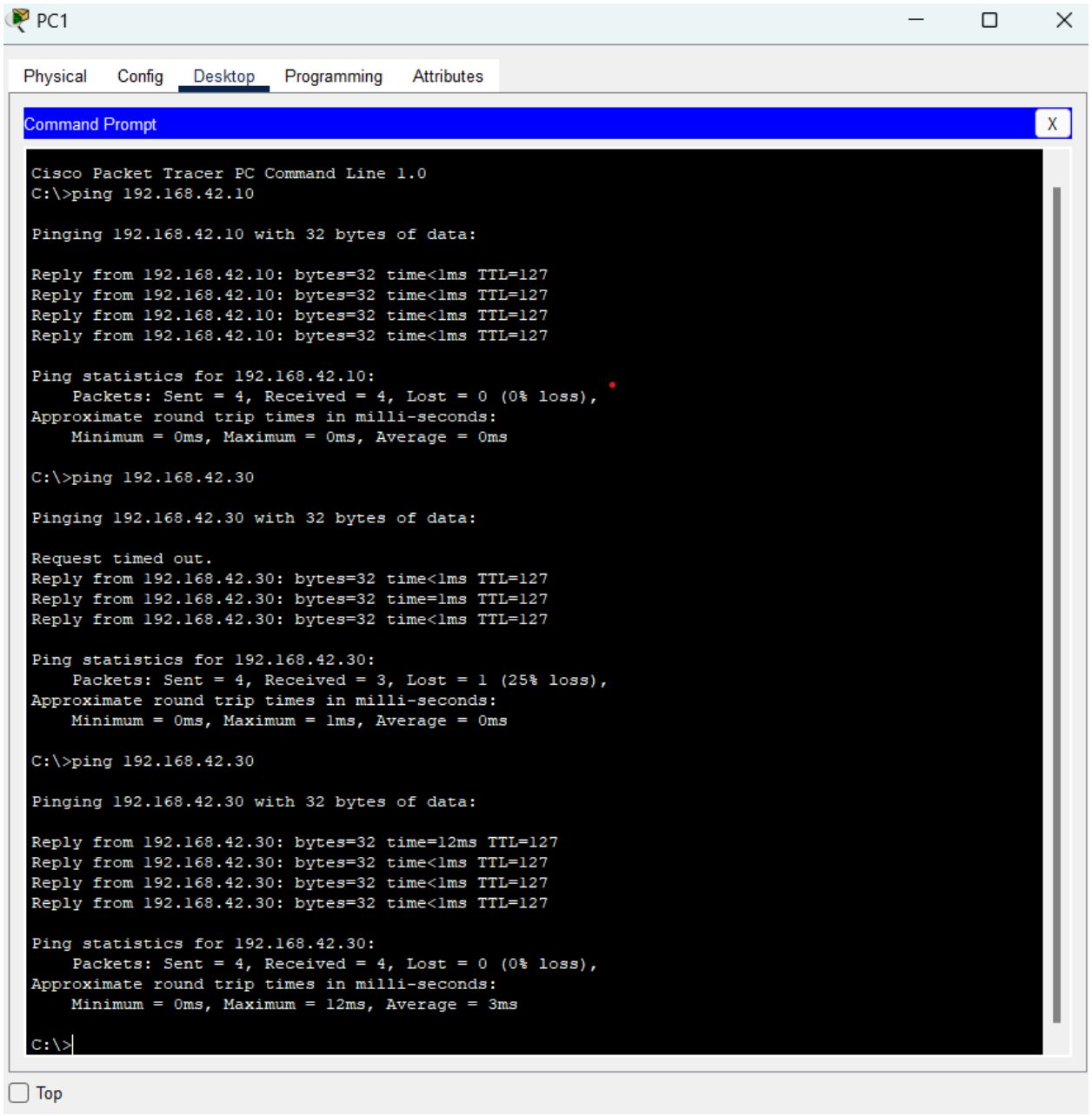
To configure router, I clicked on the router, went to CLI and wrote the following code:
132# config t
132(config)# interface gigabitEthernet 0/0
132(config-if)# ip address 196.168.32.1 255.255.255.0
132(config-if)# no shutdown
132(config-if)# exit
137(config)# interface gigabitEthernet 0/1
132(config-if)# ip address 196.168.42.1 255.255.255.0

132(config-if)# no shutdown
132(config-if)# exit



Step 4: Using the ping command from the terminal

To ping, I clicked on PC1, went to Desktop, then Command Prompt. In the command prompt I wrote: Ping 192.168.42.10 then again checked for 192.168.42.30 and got reply 4 times.



Questions (Answer to the point):

Q1. Write the command to check the status of all interfaces in a router.
Ans: show interface

Q2. Why do we use switches and not hubs?
Ans: We prefer switches over hubs because switches only direct data to specified devices, they decrease network congestion and improve security and performance.

Q3. How do you make all the configuration changes in a Cisco device persistent? What would happen if you don't do this?
Ans: We use the command - (copy running-config startup-config). Any changes I make will be lost when the device reboots if I don't save the running configuration to the starting configuration.

Q4. What are the interfaces of the router? Why are they necessary?
Ans: Numerous interface types are commonly found on routers. The following are the main interface types on routers-
1. fastEthernet
2. gigabitEthernet
3. serial
4. VLAN
Interfaces are necessary for routers because they serve as the connection points to different networks. Routers operate at the network layer (Layer 3) of the OSI model and use routing tables to make decisions about how to forward data between networks. Interfaces are how routers send and receive data on these networks. Because there are so many different types of interfaces, routers may offer a wide range of technologies and connectivity options while being flexible and adaptable to varied networking circumstances.

Q5. Why is the default gateway necessary?
Ans: As it acts as the router's exit point for traffic intended distant networks, a default gateway is required. It makes resources outside of a subnet accessible by facilitating communication between devices in a local network and those in other networks.

Challenges (if any):

While configuring the router, I needed to remember all the commands and as it was my first time, it took me a while to finish such a simple thing.